

Seminar Presentation

Contrastive Learning for Sequential Recommendation

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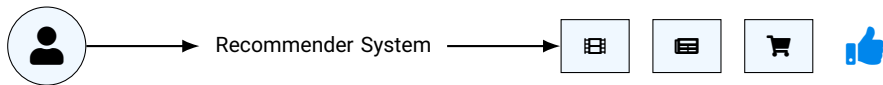
May 2025

Xu Xie et al. (2022). "Contrastive learning for sequential recommendation". In: *2022 IEEE 38th international conference on data engineering (ICDE)*. IEEE, pp. 1259–1273

Introduction – Sequential Recommendation

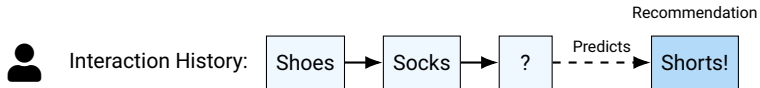
► Recommender Systems

- Help users discover relevant items (e.g., products, movies, news).
- Many common approaches assume static user preferences.



► Sequential Recommendation

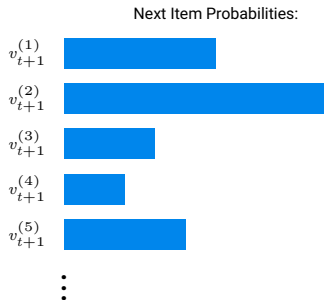
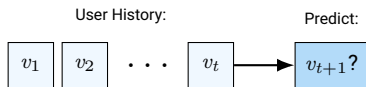
- Recommendations based on a user's ordered interaction history.
- Able to better capture dynamic user interests over time.



Introduction – Sequential Recommendation

► Formulation as a Prediction Task

- Given an ordered sequence of historical user interactions.
- Predict the next item the user will interact with.
- Very similar to the language modelling task in NLP.



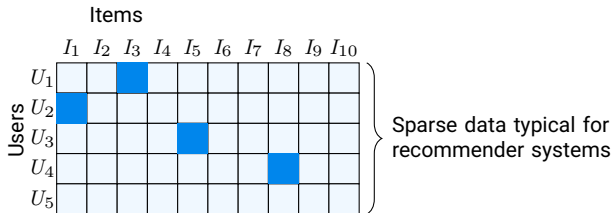
Introduction – Challenges in Sequential Recommendation

► Data Sparsity

- Most users and items have only few interactions.
- Difficult to learn meaningful patterns from short sequences.

► Learning Good User Representations

- Deep models have many parameters, need lots of data.
- Next-item prediction alone can overfit and struggle to generalize.



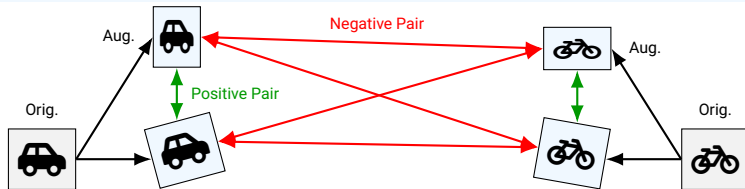
Introduction – Contrastive Learning

► Self-Supervised Learning

- Learning from the data itself without explicit human-provided labels.
- Define some pretext task based on the input data.
- Has been effective in other domain like CV and NLP.

► Contrastive Learning

- The pretext task is to compare different instances.
- The model learns to tell similar things apart from dissimilar things.
- Positive pair: Two augmented versions from same input.
- Negative pair: Augmented versions of different inputs.



CL4SRec – The Proposed Framework

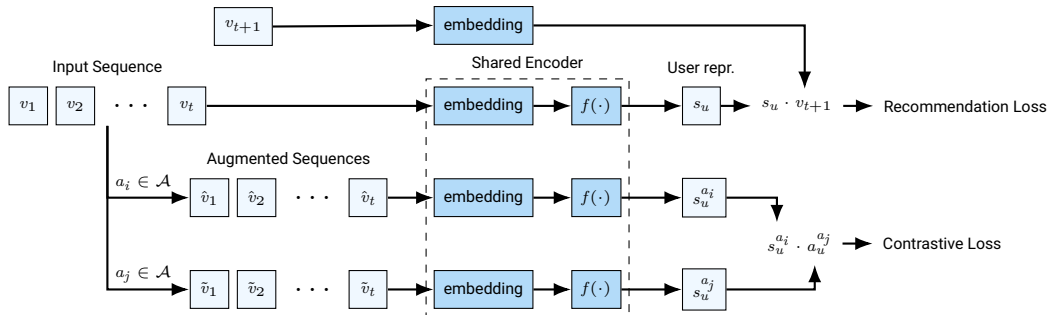
- ▶ User Representation Encoder

- Takes a user's interaction sequence as input, outputs a user representation.

- ▶ Encoder shared between two learning objectives.

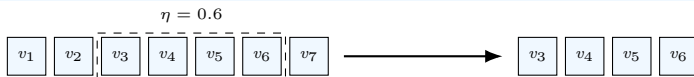
- Recommendation Loss: Standard task of predicting next item in the sequence.

- Contrastive Loss: Compare augmented views from the original interaction sequences.

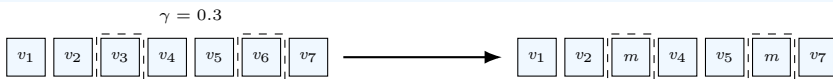


CL4SRec – Data Augmentation Strategies

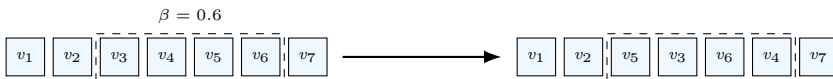
- ▶ **Item Crop** → Randomly select a continuous subsequence of the sequence.



- ▶ **Item Mask** → Randomly replace some items with a special mask item.

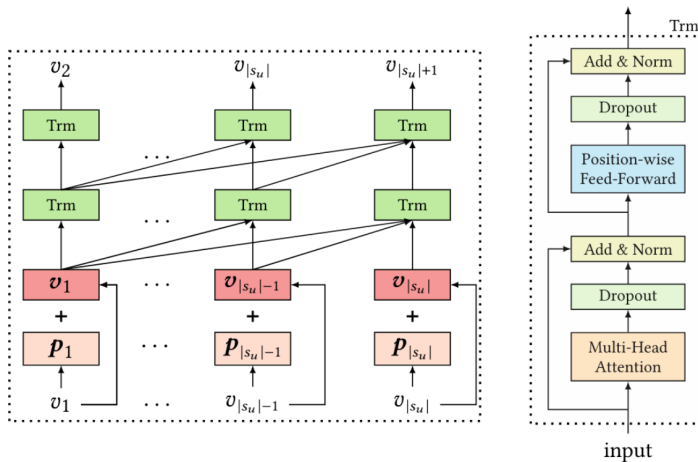


- ▶ **Item Reorder** → Randomly shuffle items within a continuous subsequence.



CL4SRec – Model Architecture

- User Representation Model using a Transformer (based on SASRec)



CL4SRec – Training Process

► Recommendation Loss

- Typically cross-entropy, predicting the true next item versus other items.
- Uses only a sampled softmax for efficiency.
- $\mathcal{L}_{\text{main}}(s_u) = -\log \frac{\exp(s_u \cdot v^+)}{\exp(s_u \cdot v^+) + \sum_{v^- \in V^-} \exp(s_u \cdot v^-)}$

► Contrastive Loss

- Aims to maximize similarity for positive pairs and minimize for negative pairs.
- $\mathcal{L}_{\text{cl}}(s_u^{a_i}, s_u^{a_j}) = -\log \frac{\exp(s_u^{a_i} \cdot s_u^{a_j})}{\exp(s_u^{a_i} \cdot s_u^{a_j}) + \sum_{s^- \in S^-} \exp(s_u^{a_i} \cdot s^-)}$

► Multi-Task Learning

- The model is trained to optimize both tasks simultaneously.
- $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{main}} + \lambda \mathcal{L}_{\text{cl}}$

Evaluation – Experimental Setup

- ▶ Four public real-world datasets.
 - Amazon Beauty, Amazon Sports, Yelp, MovieLens-1M.
 - Filtered to only users and items with at least 5 interactions.
 - Only uses item identifiers, not ratings or review text.
 - Interactions grouped by user and ordered by timestamp.
 - Last item per user for testing, Second-to-last item for validation.

Dataset	#users	#items	#actions	avg.length	density
Beauty	22,363	12,101	198,502	8.8	0.07%
Sports	25,598	18,357	296,337	8.3	0.05%
Yelp	30,983	29,227	321,087	10.3	0.04%
ML-1M	6,040	3,953	1,000,209	165.6	4.19%

- ▶ Evaluation Metrics
 - HR@k: Ratio the correct item appears in the top-k recommendations.
 - NDCG@k: Consider also the rank in the top-k at which the correct item appear.

Evaluation – Comparison With Other Methods

► Baselines for Comparison

- Non-personalized: Popularity-based
- Sequential: GRU4Rec+, SASRec

- Non-sequential: BPR-MF, NCF
- Other: GC-SAN, S³-Rec_MIP

Dataset	Metric	Pop	BPR-MF	NCF	GRU4Rec+	SASRec	GC-SAN	S3-RecMIP	CL4SRec	Improv.
Beauty	HR@10	0.0114	0.0299	0.0293	0.0399	<u>0.0630</u>	0.0444	0.0566	0.0681	8.10%
	NDCG@10	0.0053	0.0122	0.0130	0.0201	<u>0.0276</u>	0.0225	0.0270	0.0299	8.33%
Sports	HR@10	0.0090	0.0188	0.0219	0.0259	<u>0.0328</u>	0.0266	0.0298	0.0387	17.98%
	NDCG@10	0.0051	0.0083	0.0100	0.0131	<u>0.0148</u>	0.0139	0.0146	0.0171	15.54%
Yelp	HR@10	0.0095	0.0273	0.0267	0.0276	0.0300	0.0267	<u>0.0313</u>	0.0349	11.50%
	NDCG@10	0.0049	0.0121	0.0119	0.0131	0.0131	0.0136	<u>0.0157</u>	0.0171	8.92%
ML-1M	HR@10	0.0162	0.0354	0.0313	0.1806	0.1902	0.1906	<u>0.1952</u>	0.1975	1.18%
	NDCG@10	0.0079	0.0158	0.0143	0.0835	0.0904	0.0895	<u>0.0917</u>	0.0928	1.20%

- CL4SRec outperforms all baselines across all metrics and datasets.
 - Particularly effective on sparser datasets.

Evaluation – Influence of Hyper-Parameters

- ▶ → All three augmentation methods improve performance over not using CL.
- The best augmentation method varies by dataset.
- The proportion of augmentation is also an important hyperparameter.
- A well-chosen λ is crucial.

Evaluation – Data Augmentation vs. Contrastive Learning

- ▶ Full CL4SRec further improves over just applying augmentations.
 - Data Augmentation alone helps over standard SASRec.
 - Contrastive Learning on top helps more.

Evaluation – Better User Representations

- ▶ Experiment on Yelp dataset, which contains social graph of friends.
 - Compute the histogram of similarities between user representation of friends.
 - Friends have more similar representations under CL4SRec than under SASRec.
 - Suggests CL4SRec learns user representations that are “more meaningful”.

Evaluation – Strengths and Weaknesses of the Paper

► Strengths

- Novel application of contrastive learning to sequential recommendation.
- Achieves state-of-the-art results, especially on sparse data.
- Performs extensive evaluation to verify those claims.

► Weaknesses

- More theoretical understanding of why CL works so well here could be beneficial.
- They claim CL4SRec is “model-agnostic”, but only evaluated with a transformer model.
- Limited discussion about the rationale for the chosen augmentations.

Quiz

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